

Introduction

Purpose

This document specifies the requirements for CalvinFinder, a mobile app that helps users find classes, buildings and points of interest (POIs) around Calvin University. It defines what the system must be able to do and what qualities it must have so the team shares one vision for what the product should like before, during and after the development process.

Intended Audience

Developer team: Team G

Course Instructor: Prof. Victor Norman

Stakeholders: Team H

Future maintenance team

Intended Use

CalvinFinder is intended for anyone visiting or navigating Calvin University who needs to locate a building, room, or amenity. Typical uses would include an incoming student finding a classroom on the first day of classes, a Calvin guest trying to locate a building or anyone looking to find the nearest printer, bathrooms or vending machine. It is used to reference and help an individual find the way.

Product Scope

CalvinFinder is a mobile app that shows an interactive campus map, lets users select buildings to view their layouts, floors and displays POIs inside the buildings. Unlike other campus navigation tools, it combines classroom locations and POIs on one platform.

Definitions and Acronyms

Term	Definition
POI	Point of Interest such as vending machine, printer, water fountain, bathroom
Building Layout	A floor plan showing rooms and features in the building
Room	A number space in the building (e.g., SB 010 in Science Building)
Calvin ID	A Calvin University user account number
Guest	Someone using the app without signing in
Admin	A person who maintains the data
SRS	Software Requirements Specification
UI	User Interface

MVP	Minimum Viable Product
FR/NFR	Functional / Non-functional requirement

Overall Description

Calvin Finder is a campus navigation app for Calvin University.

User Needs

Primary users: incoming students. These are newly admitted or first-year students who are enrolled members of the university but have little familiarity with the campus. They use the app to find classrooms, residence halls, dining locations, and offices, and to learn the layout over their first weeks. Getting lost makes them late, adds stress to an already stressful transition, and discourages them from asking for help.

Secondary users: visitors. These are external users with no formal role in the university, such as visiting parents and prospective students touring campus. They spend a short time on campus and use the app to reach specific destinations such as admissions, parking, and event venues without needing a guide. Getting lost costs them time and gives a poor first impression of the university.

Dependencies. Calvin Finder is a mobile app and depends on the following to function:

- **Mobile framework:** React Native, and the iOS and Android versions it supports.
- **Backend:** Supabase for the database and data storage, which requires an active Supabase project and network access to it.
- **Campus data:** Floor plans provided by Calvin University in PDF format. The team will manually convert these into nodes, connections, and room labels for routing, so the app depends on the plans being accurate and on the team completing this conversion.
- **Third-party libraries:** React Native packages for map rendering, navigation and routing, and UI, and their continued compatibility with the framework version used.
- **Internet connectivity:** The app requires an internet connection (Wi-Fi or cellular data) to retrieve campus data and does not support offline use.
- **Reused components:** No software is reused from previous projects.

Assumptions.

- The app does not use GPS or real-time location tracking. Users select their starting location manually, and routes are computed from that point.
- Users have a smartphone with an active internet connection while using the app.
- Calvin's floor plans are accurate at release, and campus changes are infrequent enough to be handled by manual updates.
- Supabase remains available and its usage limits are sufficient for the project's expected load.

Systems Functions and Requirements

Functional Requirements

- FR-1 Display campus map. The system shall display an interactive campus map on launch.
- FR-2 Pan and zoom. The user shall be able to pan and zoom the map.
- FR-3 Select building. The user shall be able to tap a building to view its details.
- FR-4 View layout. The system shall show a selected building's floor layouts.
- FR-5 Switch floors. For multi-story buildings, the user shall be able to switch between floors.
- FR-6 Room information. The layout shall display room numbers and key features (entrances, stairs, elevators, restrooms).
- FR-7 Display POIs. The system shall display POI markers on the campus map and on building layouts.
- FR-8 POI details. Selecting a POI shall show its name, category, and location.
- FR-9 Filter POIs. The user shall be able to filter POIs by category.
- FR-10 Search. The user shall be able to search buildings, rooms, and POIs by name or number, with suggestions as they type.
- FR-11 Location list. The system shall provide a list of buildings that expand to show their rooms.
- FR-12 List-to-map link. Selecting a list item shall show that location on the map.
- FR-13 Sign in and guest access. The user shall be able to sign in with a Calvin ID or continue as a guest.
- FR-14 Password recovery. Signed-in users shall be able to reset a forgotten password.
- FR-15 Favorites. The user shall be able to save, view, and remove favorite locations.
- FR-16 Settings. The user shall be able to adjust display and permission settings.
- FR-17 Report an issue. The user shall be able to report a missing or incorrect location.
- FR-18 Manage locations. An admin shall be able to add, edit, and delete buildings, rooms, POIs, and layouts.
- FR-19 Error handling. The system shall show clear messages and a retry option when data fails to load, the user is offline, or data is missing.

EARS Format

ID	Statement
FR-1	When the app launches, the system shall display the interactive campus map.
FR-2	When the user performs a pinch or drag gesture on the map, the system shall zoom or pan accordingly.
FR-3	When the user taps a building on the map, the system shall open that building's details.
FR-4	When the user opens a building's details, the system shall display its floor layouts.

FR-5	When the user selects a different floor, the system shall display that floor's layout.
FR-6	When a floor layout is displayed, the system shall show room numbers and key features on it.
FR-7	When the map or a layout is displayed, the system shall show POI markers on it.
FR-8	When the user taps a POI marker, the system shall display that POI's name, category, and location.
FR-9	When the user selects one or more POI categories, the system shall show only matching POIs on the map.
FR-10	When the user types in the search field, the system shall return matching buildings, rooms, and POIs with suggestions.
FR-11	When the user expands a building in the location list, the system shall show that building's rooms.
FR-12	When the user taps a location list item, the system shall center the map on that location.
FR-13	When the user reaches the sign-in screen, the system shall allow sign-in with a Calvin ID or continuing as a guest.
FR-14	When a signed-in user requests a password reset, the system shall send reset instructions to their account.
FR-15	When the user favorites a location, the system shall add it to their favorites list and make it removable.
FR-16	When the user opens settings, the system shall allow the user to adjust display and permission settings.
FR-17	When the user submits a location issue report, the system shall record it for admin review.
FR-18	When an admin submits a location change, the system shall update the stored building, room, POI, or layout data.
FR-19	When a data load fails or the device is offline, the system shall display an error message with a retry option.

Non-functional Requirements

- NFR-1 The campus map shall load within 3 seconds on a typical mobile connection.
- NFR-2 Search results shall appear within 1 second of the user's query.
- NFR-3 Switching floors or filtering POIs shall update the display within 1 second.
- NFR-4 Only authenticated users shall be able to access admin functions.
- NFR-5 Passwords shall never be stored in plain text; the system shall store only salted hashes or delegate authentication to campus login.
- NFR-6 Data in transit between the app and server shall be encrypted.

- NFR-7 A signed-in user's session shall expire after a period of inactivity.
- NFR-8 A first-time user shall be able to find a building without instructions.
- NFR-9 The app shall remain usable, with cached data available, when the network is unavailable.
- NFR-10 The app shall use device location only with the user's permission and shall not share it, in line with campus data-privacy expectations.
- NFR-11 The app shall run on current and previous major iOS and Android versions and adapt to different screen sizes.

System Features

Campus Map, Building Layouts, POIs, Search and Browse, Accounts and Favorites, Feedback & Admin, Error handling

Other Requirements

- **Database will hold:**
 - Map layouts
 - Names of rooms
 - POIs
 - User schedules
 - Routes for pathing

Relationships: A building has many floors, a Floor has many rooms, A POI belongs to one category and may belong to a Building or Floor. A user has many favorites and may submit many issue reports.